



COMPUTER ARCHITECTURE: INTEGRATING CLOUD ARCHITECTURE



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ABSTRACT

This project redesigns an undergraduate Computer Architecture course to connect foundational architecture concepts with modern cloud computing, high-performance computing, and artificial intelligence workloads. The course uses Science Gateway resources, with Jetstream2 as a key experiential platform, to help students move from textbook models of processors, memory, storage, I/O, and parallelism to the virtualized and distributed architectures they encounter in contemporary cloud systems. Students investigate how traditional architecture maps to cloud resources, how AI workloads have driven the evolution of GPUs and specialized accelerators, and how architecture decisions affect performance, scalability, cost, and resource utilization. The redesign is informed by the instructor's professional experience in data architecture, data engineering, cloud computing, data security, healthcare data, enterprise architecture, and application modernization. Students use Jetstream2 or comparable Science Gateway resources to provision and analyze computing environments, examine CPU and memory resources, explore storage and networking, and relate these resources to IaaS, PaaS, and SaaS models. A culminating architecture project asks students to select an AI or data-intensive workload, identify its processor, memory, storage, parallel-processing, and accelerator requirements, and propose an appropriate on-premises or cloud architecture. The project addresses a common instructional gap: students may understand individual hardware components without seeing how those components become scalable cloud and AI infrastructure. The redesign therefore emphasizes hands-on experimentation, architecture comparison, and industry-relevant design thinking. A key lesson is that cloud and AI should not be taught as separate topics from computer architecture; they are contemporary expressions of the same architectural principles.

INTRODUCTION

Computer Architecture is an undergraduate computing course designed to establish a foundation in how computer systems execute workloads and how architecture evolves to support modern applications. The redesign addresses the gap between traditional architecture framework and the cloud, AI, and data-intensive systems students are likely to encounter in industry. Students connect processors, memory, storage, I/O, parallelism, and performance to virtualization, cloud infrastructure, GPUs, AI accelerators, and distributed computing. Jetstream2 provides an experiential environment in which students can investigate these relationships rather than treating cloud infrastructure as an abstraction. The course also draws on the instructor's professional experience in enterprise data architecture, cloud platforms, security, healthcare data, and modernization.

Course Goals

Master foundational computer architecture concepts: CPU, memory hierarchy, storage, I/O, pipelining, and performance
Develop systems thinking by mapping physical architecture concepts to virtualized and distributed cloud environments
Emphasize professional practice through architecture documentation, reproducibility, analysis, and communication

- Use Jetstream2/Science Gateway resources to connect architecture theory to cloud computing and AI workloads

Grade Computation

Architecture Design Project using JetStream	15%
Cloud / Jetstream2 Virtual Labs	15%
Assignments and Discussion	30%
Attendance	10%
Fina Exam	30%

Mentor Feedback Incorporated

Mentor feedback to incorporate: strengthen the explicit mapping between Jetstream2 resources and CPU, memory, storage, I/O, and networking concepts.
Mentor feedback to incorporate: make the AI-to-architecture connection concrete through GPU/accelerator and AI workload examples.

- Mentor feedback to incorporate: individual project using JetStream2

IMPACT ON STUDENTS

The primary users are undergraduate computer science students. The redesign gives students practical exposure to cloud and research-computing resources while reinforcing foundational architecture concepts. Students gain experience reasoning across hardware, operating systems, virtualization, cloud services, and AI infrastructure. The goal is to increase confidence in analyzing real computing systems and to make architecture more relevant to contemporary industry environments.

- What worked: framing Jetstream2 as a bridge from familiar CPU/memory/storage concepts to cloud infrastructure makes abstract cloud concepts more concrete.
- What we'd change: provide more guided architecture-to-resource mapping examples before students begin independent experiments.
- What surprised us: AI workloads make architecture concepts immediately relevant because students can see why parallelism, memory bandwidth, and accelerators matter.

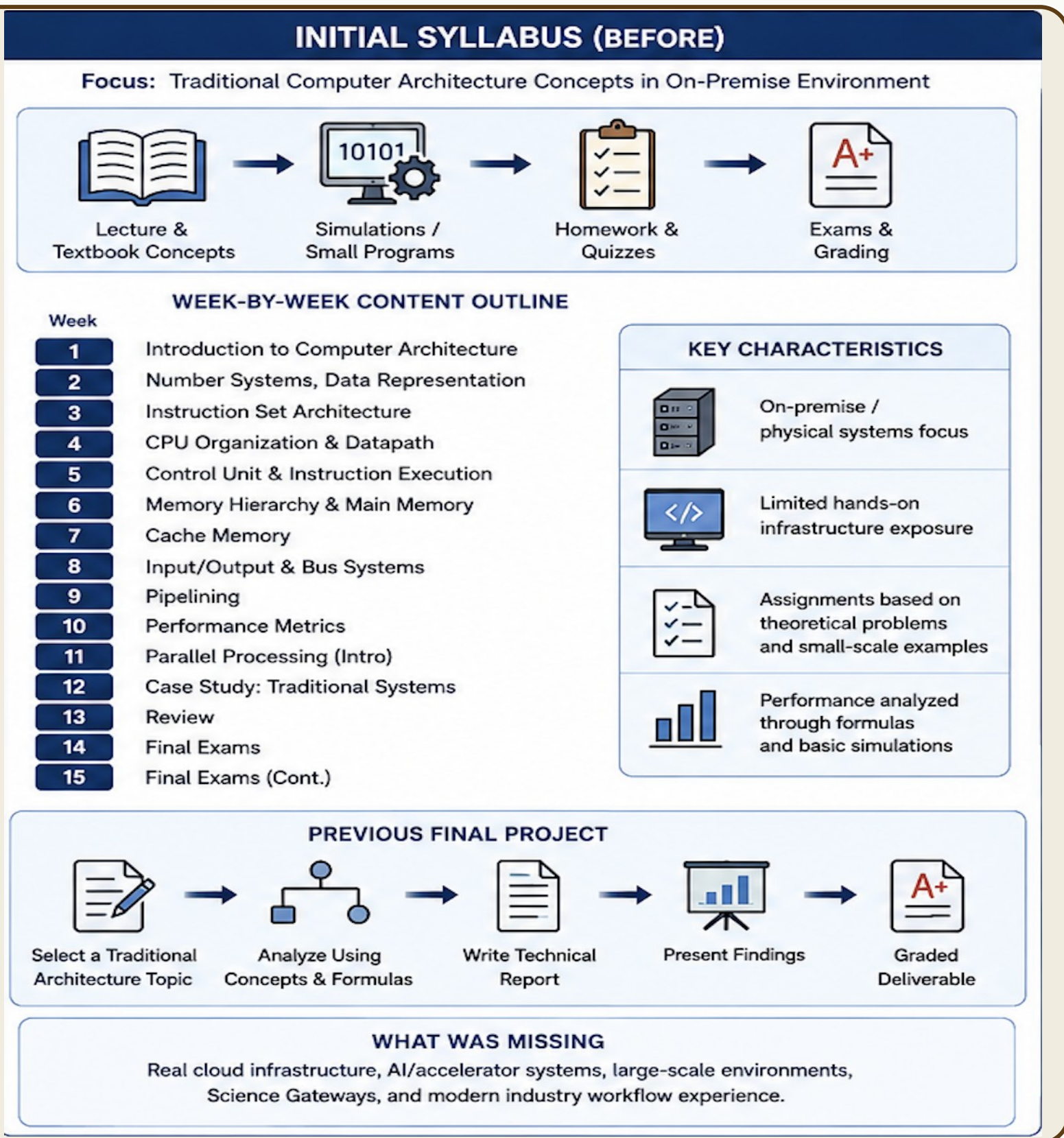
COURSE REDESIGN: BEFORE & AFTER

INITIAL SYLLABUS

- Traditional focus on CPU, memory, storage, I/O, pipelining, and performance.

- Students primarily learned architecture through lectures, textbook examples, and conventional assignments.

- Limited opportunity to connect physical architecture to cloud infrastructure, AI accelerators, and scalable workloads.



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TARGET COURSE DESCRIPTION

Course Name	Computer Architecture
Course Number	CMP 334
Department	Science, Technology, Health, and Human Services
Enrollment	15
Prerequisites	Introductory Programming / Computer Science Foundations
Project Repo	https://github.com/gestureb-cpu/Course-Syllabus

Course Description

An undergraduate course covering the organization and operation of computer systems, including processors, instruction execution, memory hierarchy, storage, I/O, performance, parallel computing, and modern system design. The course extends these foundations into cloud computing and artificial intelligence, examining virtualization, IaaS/PaaS/SaaS, GPUs, specialized AI accelerators, and distributed infrastructure for AI and data-intensive workloads.

Key Content by Week

Module 1 — Introduction to Computer Architecture & Cloud Computing
Module 2 — Computer System Overview, Operating Systems & Virtualization
Module 3 — Performance, Pipelining & Architectural Metrics
Module 4 — Memory Hierarchy, Caching & AI Workloads
Module 5 — I/O Systems, Storage & Data Pipelines
Module 6 — Parallel Computing, GPUs & AI Acceleration
Module 7 — Distributed Systems & Cloud Computing — Jetstream2 integration
Module 8 — Emerging Architectures & Capstone AI System Design

SCIENCE GATEWAY RESOURCES USED

Jetstream2 is the primary Science Gateway/cloud resource for the course. Students use its research-computing environment to explore virtual machines, CPU and memory allocation, storage, networking, operating systems, resource provisioning, and workload execution. It fits the course because it provides a practical bridge between physical computer architecture and cloud abstractions. The course will also investigate other Science Gateway AI/HPC resources that can provide access to GPUs, accelerators, parallel computing, and scalable AI workloads.

<https://hairpilot.org/> - Cloudbank
<https://docs.jetstream-cloud.org/>

SCIENCE GATEWAY GOAL

Science Gateway Goal:

My goal is to leverage Jetstream2 to provide students with cloud-based access to real CPU, GPU, memory, storage, and networking resources, allowing them to **experiment with and measure how computer architecture components affect AI workload performance**

Jetstream2 Architecture Mapping Workflow
Physical CPU / Memory / Storage / Network → Virtual Resources → Cloud Workload → AI/Parallel Workload Analysis

FINAL PROJECT / DELIVERABLE

Computer & Cloud Architecture Design Project

- Select an AI or data-intensive workload and define its computational requirements.
- Identify CPU, memory, storage, networking, parallelism, and accelerator requirements.
- Provision or analyze a suitable cloud/Science Gateway environment such as Jetstream2.
- Compare an on-premises architecture with an IaaS/PaaS/SaaS alternative.
- Produce an architecture diagram, technical rationale, performance/scalability analysis, and presentation.

Workload → Requirements → Architecture Mapping → Jetstream2/Cloud Experiment → Performance & Scalability Analysis → Architecture Design → Presentation

Project Goals

- Demonstrate understanding of computer architecture and organization
- Map traditional architecture concepts to cloud computing resources
- Analyze architecture requirements for AI workloads, including GPUs and specialized accelerators
- Evaluate on-premises versus IaaS, PaaS, and SaaS architectures

EXPANSION IDEAS

- Expand the course with additional GPU-enabled Science Gateway resources and AI datasets.
- Create a follow-on cloud architecture/HPC/AI systems module or workshop using reproducible cloud labs.
- Develop collaborations with campus research computing, data science, cybersecurity, and AI programs and industry partners.

REFERENCES

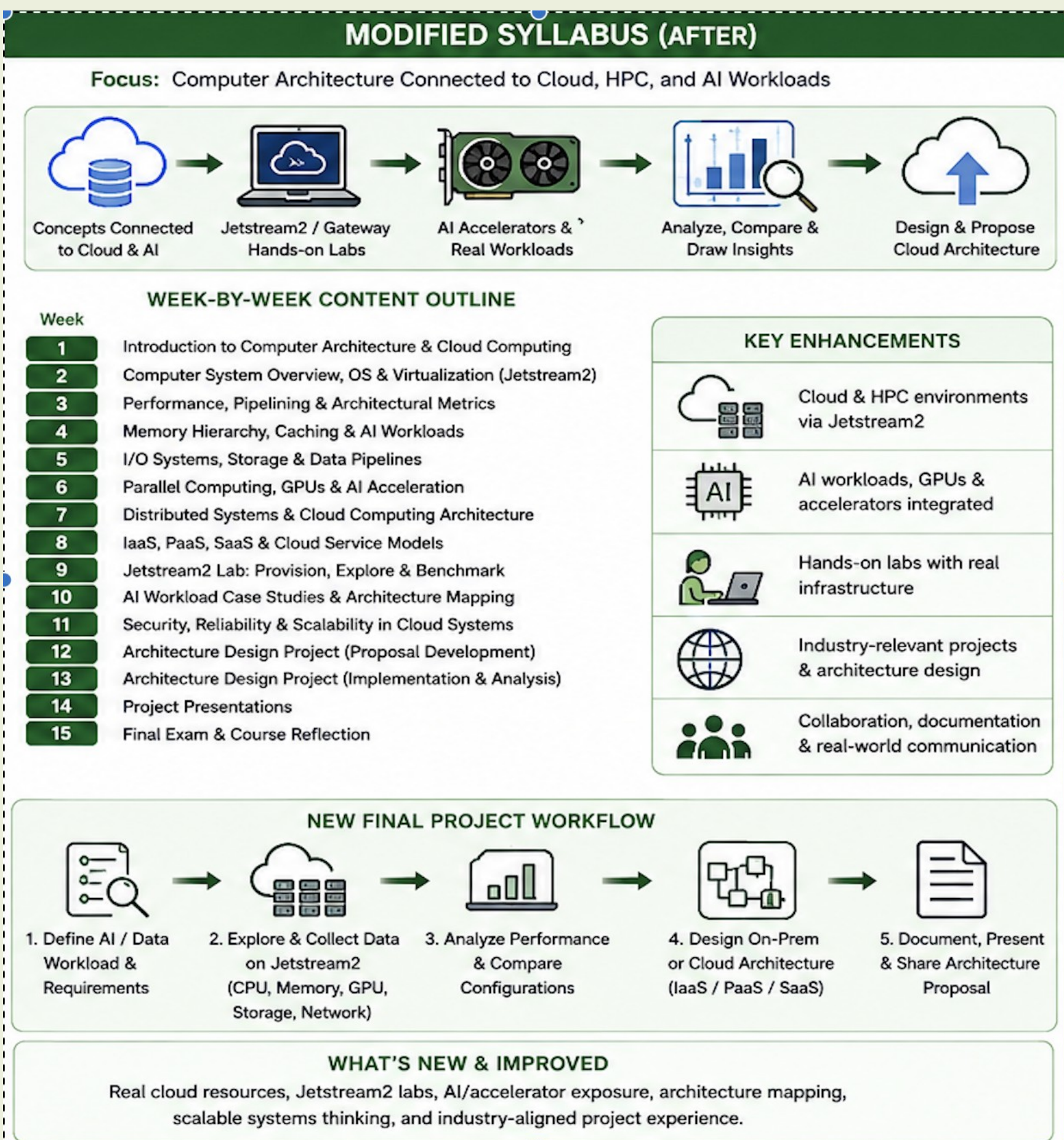
- David A Patterson, John L. Hennessy – textbook/reference Computer Organization and Design
- Jetstream2, National Science Foundation-supported research computing infrastructure.
- FacultyHack@Gateways 2026 resources and project guidance.

LESSONS LEARNED

- The redesign is intentionally iterative. Future offerings will use student performance and feedback to refine workload selection, lab instructions, resource provisioning guidance, and assessment. A major continuing focus will be balancing architectural depth with the practical complexity of cloud systems.

MODIFIED SYLLABUS

- Adds cloud architecture, virtualization, AI workloads, GPUs, accelerators, and distributed systems.
- Jetstream2 is introduced in the cloud/distributed computing portion and used for experiential investigation.
- Assessment adds cloud labs, AI architecture analysis, and a final architecture design project.



PROJECT GITHUB REPO

<https://github.com/gestureb-cpu/Course-Syllabus>

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